

PHALGUNI NANDA

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RESEARCH INTERESTS

My research develops rigorous non-asymptotic analyses that characterize sample complexity and learning dynamics in reinforcement learning, with the goal of designing algorithms that are practically effective and provably efficient in high-dimensional stochastic settings for sequential decision making.

EDUCATION

Purdue University

Ph.D. in Operations Research

Edwardson School of Industrial Engineering

Advisor: Prof. Zaiwei Chen

M.S. in Mathematical Statistics

Department of Statistics

West Lafayette, IN, USA

May 2025 – Present

August 2022 – May 2025

National Institute of Technology Rourkela

Integrated (Five-Year) M.Sc. in Mathematics

Dissertation: Stochastic comparisons of series and parallel systems with heterogeneous components

Advisor: Prof. Suchandan Kayal

Rourkela, Odisha, India

July 2013 – June 2018

HONORS AND AWARDS

1. Best Paper Runner-Up at ACM SIGMETRICS 2026
2. Student Travel Grant to present at ACM SIGMETRICS 2026
3. Nominated by the Purdue Industrial Engineering Graduate Committee to present in the IISE Doctoral Colloquium at the IISE Annual Conference and Expo 2026
4. Scholarship for Higher Education (SHE) under INSPIRE programme for the duration of my Five-Year M.Sc. program

PUBLICATIONS

Doctoral Research

Submitted/Working Papers

1. P. Nanda and Z. Chen. Natural policy gradient as doubly smoothed policy iteration: A Bellman-operator framework. *Submitted to a Machine Learning Conference.*
2. Z. Chen and P. Nanda. From set convergence to pointwise convergence: finite-time guarantees for average-reward Q-learning with adaptive stepsizes. *Under Review at Mathematics of Operations Research.*

Published Articles

1. P. Nanda and Z. Chen. A minimal-assumption analysis of Q-learning with time-varying policies. *Proceedings of the ACM on Measurement and Analysis of Computing Systems*, 10(2) : 1–43, 2026,

Selected Pre-Doctoral Research

1. G. M. M. Reddy, P. Nanda, M. Vynnycky. A compositional approach for two-dimensional, two-phase, nonlinear inverse Stefan problems using the method of fundamental solutions. *Studies in Applied Mathematics*, 156:e70184, 2026, <https://doi.org/10.1111/sapm.70184>
2. P. Nanda, G. M. M. Reddy. Efficient numerical solution of one-phase linear inverse Stefan and Cauchy-Stefan problems in two-dimensions: *a posteriori* error control. *Studies in Applied Mathematics*, 148(4) : 1563 – 1585, 2022, <https://doi.org/10.1111/sapm.12484>
3. P. Nanda, P. Bhuyan and A. Dewanji. Optimal replacement policy under cumulative damage model and strength degradation with applications. *Annals of Operations Research*, <https://doi.org/10.1007/s10479-021-04080-6>.

PRESENTATIONS

1. *Invited talk on Natural policy gradient as doubly smoothed policy iteration: A Bellman-operator framework* in the session Foundations of Reinforcement Learning at the 2026 INFORMS Annual Meeting, November 1–4, 2026, San Francisco, CA, USA
2. Three-Minute Thesis Pitch at the IISE Doctoral Colloquium of the IISE Annual Conference and Expo, May 15–19, 2026, Arlington, TX, USA
3. Oral presentation on *A minimal-assumption analysis of Q-learning with time-varying policies* in the session Theoretical Machine Learning at ACM SIGMETRICS, June 8–12, Ann Arbor, MI, USA
4. Poster on *Natural policy gradient as doubly smoothed policy iteration: A Bellman-operator framework* at the Stochastic Networks Conference, June 15–19, 2026, Chicago, IL, USA
5. Poster on *A minimal-assumption analysis of Q-learning with time-varying policies* at the 12th Midwest Workshop on Control and Game Theory, April 25–26, 2026, Purdue University, IN, USA
6. Poster on *Optimal replacement policy under cumulative damage model and strength degradation with applications* at the 2017 IISA International Conference on Statistics, Hyderabad, India

TEACHING

Purdue University, West Lafayette.

Teaching Assistant, Edwardson School of Industrial Engineering.

- IE 230 - Probability and Statistics in Engineering I, Spring 2026
- IE 533 - Industrial Applications of Statistics, Spring 2026

Teaching Assistant, Department of Statistics.

- STAT 113 - Statistics and Society, Fall 2022, Spring & Fall 2023, Spring & Fall 2024
- STAT 30301 - Probability and Statistics for Business, Spring 2025
- STAT 416 - Introduction to Probability, Fall 2022
- STAT 517 - Statistical Inference, Spring 2023, Fall 2023
- STAT 511 - Statistical Methods, Summer 2023, Summer 2024
- STAT 516 - Basic Probability and Applications, Spring 2024

WORK EXPERIENCE

Birla Institute of Technology and Science, Pilani - Hyderabad

Project Assistant, Department of Mathematics

Hyderabad, India

March 2020 - January 2022

Project: Adaptive and efficient method of fundamental solutions for the numerical reconstruction of boundary data in two-phase inverse Stefan problems

Principal Investigator: Prof. Gujji Murali Mohan Reddy

Funding: Anusandhan National Research Foundation (ANRF)

Indian Statistical Institute Kolkata

Summer Intern, Applied Statistics Unit

Kolkata, India

May 2017 - July 2017

Supervisors: Prof. Anup Dewanji and Prof. Prajamitra Bhuyan

PROGRAMMING

- MATLAB, R, C++, Python